CYCLES · DEEP EXR · VFX PIPELINE

MattRM2 VFX Build

Documentation — based on Blender 5.1.2

A custom Blender build for professional VFX pipelines. This document covers the production features added on top of official Blender 5.1.2, plus the included bug fixes.

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Release 1.0

0 Table of Contents

1 Deep EXR — Cycles (CPU + GPU)

Industry-standard deep compositing, Arnold / RenderMan architecture

This build adds a full **Deep EXR** implementation to Cycles, following the **Arnold / RenderMan architecture**, on **CPU, CUDA and OptiX (SVM + OSL)**. Deep EXR is the industry-standard format for high-quality depth compositing — it stores multiple colour and depth samples per pixel, which lets a compositor merge elements by true scene depth instead of a flat 2D over.

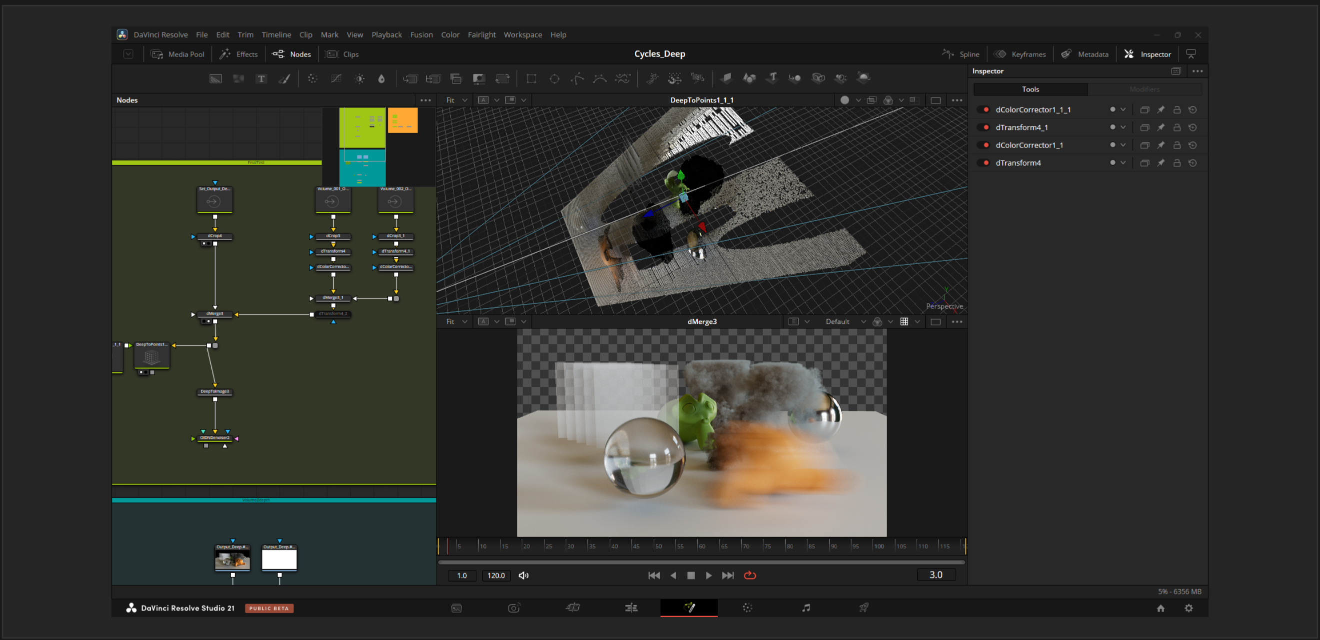


Fig. 1.1 — Cycles Deep EXR composited in DaVinci Resolve — Fusion

1.1 Overview

Key characteristics:

- **Surfaces** — recorded per sample, so Motion Blur and Depth of Field are preserved naturally. The deep flatten is pixel-perfect identical to the standard flat EXR render.
- **Volumes** — Arnold-style raymarch, decoupled from the path-traced volume integration, with **Depth of Field and Motion Blur** support. Multiple overlapping/intersecting volumes on the same view layer are fully supported.
- **Shadow catcher** — the catcher is recorded in the deep as a **shadow matte** (black RGB, shadow-density alpha), ready to deep-merge over footage — including catchers seen behind or through volumes.
- **Output** — standard OpenEXR deep scanline format, readable by Nuke, Fusion, Resolve, and any OpenEXR-compliant compositor. Single-layer deep files carry the view layer name as their part name.
- **Per-view-layer output** — each view layer gets its own subfolder and filename prefix automatically.

> **Pixel-perfect flatten**
> *The deep flatten matches the beauty render exactly: you can drop the deep EXR into your comp and get the same image as the flat render, then gain depth-based merging on top.*

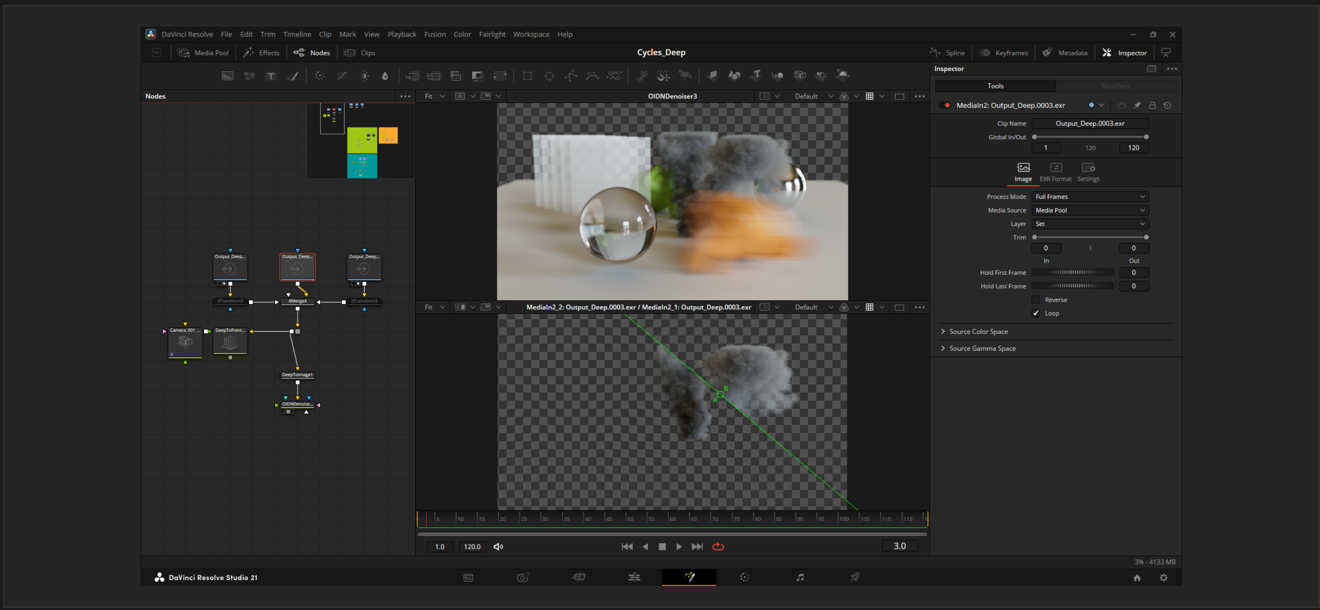


Fig. 1.2 — Deep volumes with Depth of Field, Motion Blur and shadows

1.2 Panel Reference — View Layer > Passes > Deep EXR

All Deep EXR settings live in **View Layer Properties > Passes > Deep EXR**. Each view layer has its own independent settings.

Setting	Description
Enable	Activates Deep EXR output for this view layer.
Max Depth	Maximum deep layers per pixel (surface + volume). -1 = unlimited on CPU; on GPU -1 maps to 96 — raise it for very dense volumes.
Z Threshold	Minimum Z distance to merge nearby surface fragments (scene units).
Volume Step Size	Raymarch step size for volume slabs (scene units). Match it to your volume scale — e.g. 0.01 m works well for smoke at 1 m scale.
Transparent vs World	How transparent / alpha surfaces interact with the world background in the deep.
Cache Path	Temporary directory for the per-tile deep fragment cache during rendering. Fragments are streamed to disk after each tile to keep RAM usage flat regardless of scene complexity. A local SSD is strongly recommended.
Color Depth	Float Full (32-bit) or Float Half (16-bit) per channel.
Codec	EXR compression for the deep file. Deep OpenEXR only allows ZIPS (default, recommended), RLE, or None.
Override Output Path	Custom output path template (supports // and #### frame notation).

Setting	Description
Volume DOF/MB Samples	Sub-ray count for Depth of Field and Motion Blur on the volume raymarch (in Sampling > Render). Default 16.

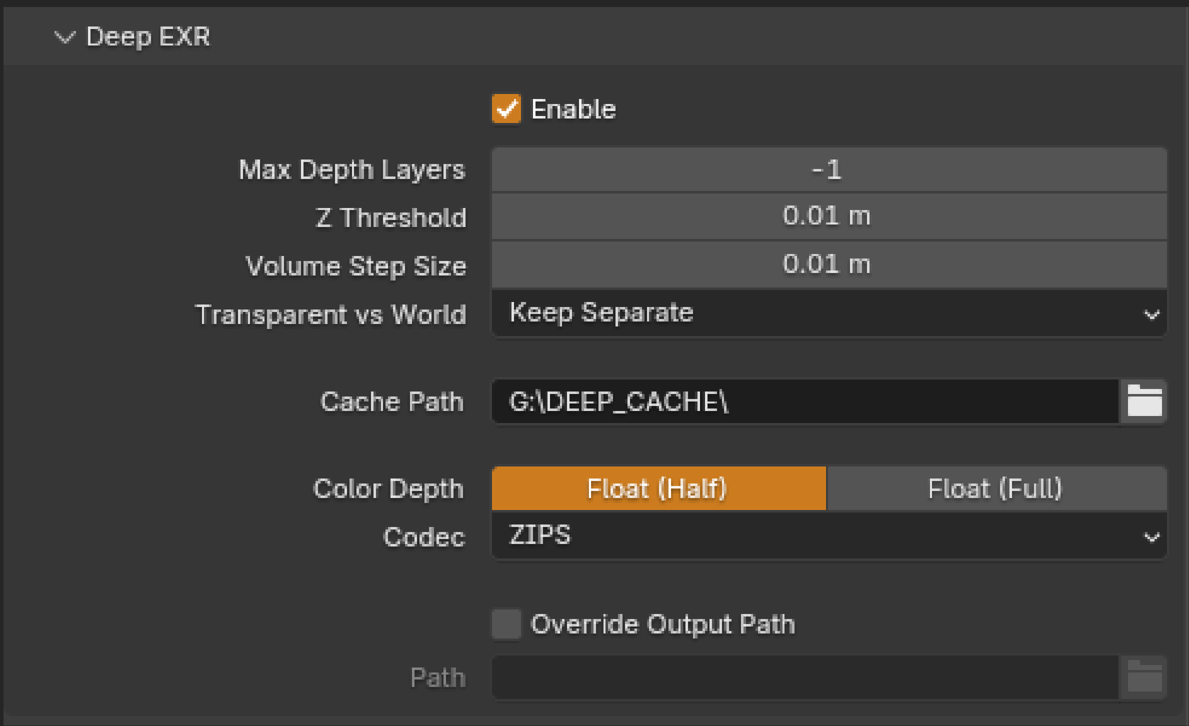


Fig. 1.3 — The Deep EXR panel in View Layer Properties

1.3 Quick Start

Follow these steps to produce your first deep EXR:

1. Set the render engine to **Cycles**. For GPU rendering, pick your device (OptiX / CUDA) in **Render Properties > Device**.
2. In **View Layer Properties > Passes > Deep EXR**, check **Enable**.
3. Set a **Cache Path** (e.g. "C:/DEEP_CACHE/" or "//DEEP_CACHE") — it must be writable; a local SSD is recommended.
4. Adjust **Volume Step Size** to match your scene scale.
5. Set your output path, or use **Override Output Path** for a custom location.

6 Render — the deep EXR is written after the render completes.

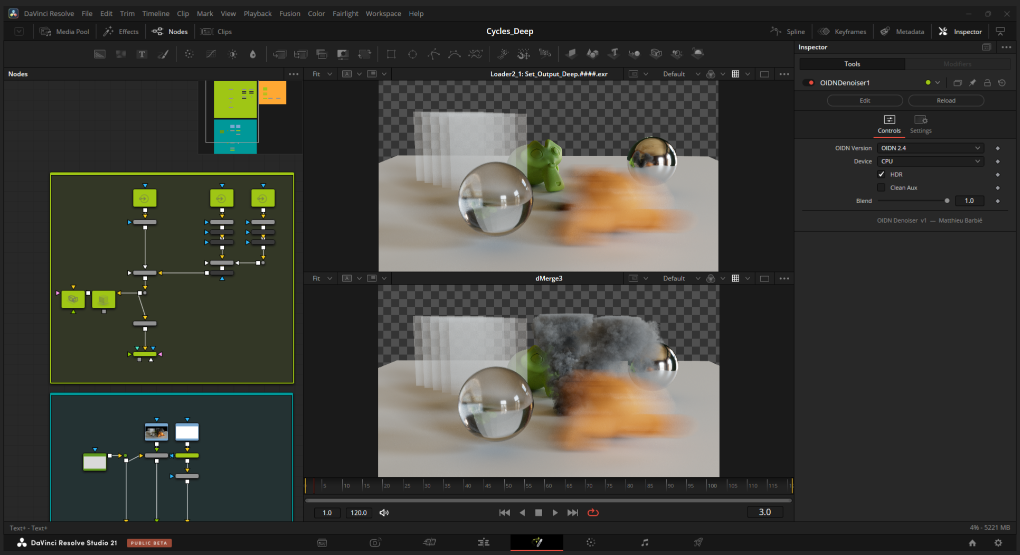


Fig. 1.4 — Basic deep compositing from the resulting deep EXR

1.4 Output Structure

Each view layer is written to its own subfolder with its own filename prefix, so multi-layer setups stay organised automatically:

```
/
 Deep/
  View_Layer/
   View_Layer_output.0001.exr
  FX_Layer/
   FX_Layer_output.0001.exr
```

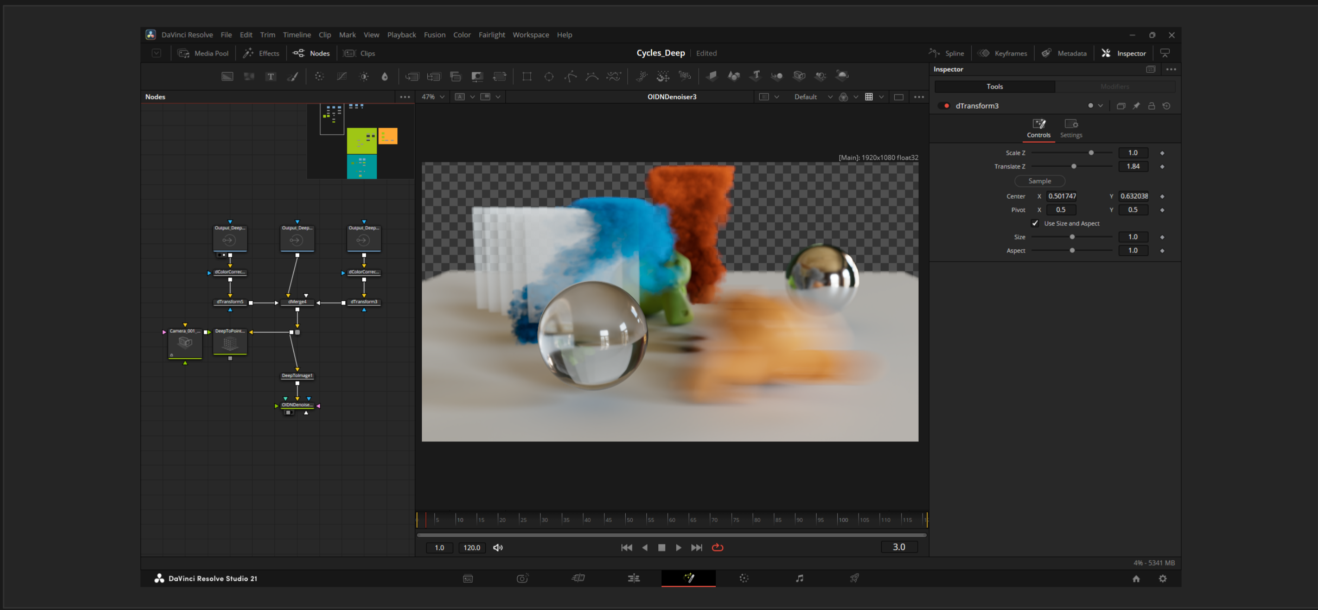


Fig. 1.5 — Volumes at different colours and depths, composited in Fusion

1.5 GPU Rendering & Volume Modes

Deep EXR runs on **CPU, CUDA and OptiX**, with both SVM and OSL volume shaders on GPU. For GPU rendering, select your device in **Render Properties > Device**.

Biased and Unbiased both supported

*Both volume integration modes work — **Unbiased** (delta-tracking) and **Biased** (ray-marching) — with no special setup for the deep.*

GPU per-pixel cap

*GPU stores deep layers in a fixed per-pixel budget: **Max Depth** defaults to 96 on GPU (the -1 sentinel), while CPU is unlimited. Raise Max Depth for very dense volumes.*

1.6 Multi-Layer Deep EXR Output

A dedicated **output format** that writes every view layer's deep data into a single multi-part OpenEXR — **one deep part per view layer** — instead of one deep file per layer in subfolders. Select it in **Output Properties > Media Type > Multi-Layer Deep EXR**, right next to *Multi-Layer EXR*.

- **One deep part per layer** — each view layer becomes its own deep part (RGBA + Z + ZBack), named after the layer. No per-layer subfolders.
- **A true deep file** — the file contains only deep parts; the beauty is the deep flatten (DeepToImage).
- **Single compression** — one codec (None / ZIPS / RLE) for the whole file, set in the Output panel; these are the only codecs valid for deep data.
- **Per-layer stats** — cycles.<layer>.samples / render_time are written per layer, just like a flat multi-layer EXR.

- **Automatic** — selecting the format enables deep recording for the render.

Compositor compatibility — Fusion / Resolve vs Nuke

Blackmagic Fusion / DaVinci Resolve *read every deep part — all layers from one file, a real convenience there.* **Nuke** *reads only the* **first** *deep part (its DeepRead handles one part per read); for isolated per-layer deep in Nuke, use the standard output instead, which writes one deep file per layer in Deep/<layer>/.*

Deep file vs AOVs

Need the flat passes / AOVs (Diffuse, Normal, Cryptomatte…) too? Render them in parallel as a standard Multi-Layer EXR — the usual VFX split of a deep file and an AOV/beauty render. Choosing any non-deep format keeps the per-layer behaviour from section 1.4 (flat render plus per-layer deep files in Deep/<layer>/ subfolders).

Roadmap — deep AOVs / DeepID

Embedding the passes inside the deep (per-fragment Diffuse, IDs…), and single-file per-layer isolation in Nuke, require per-fragment recording — that is the upcoming DeepID feature. This deep-only format is the foundation it will build on.

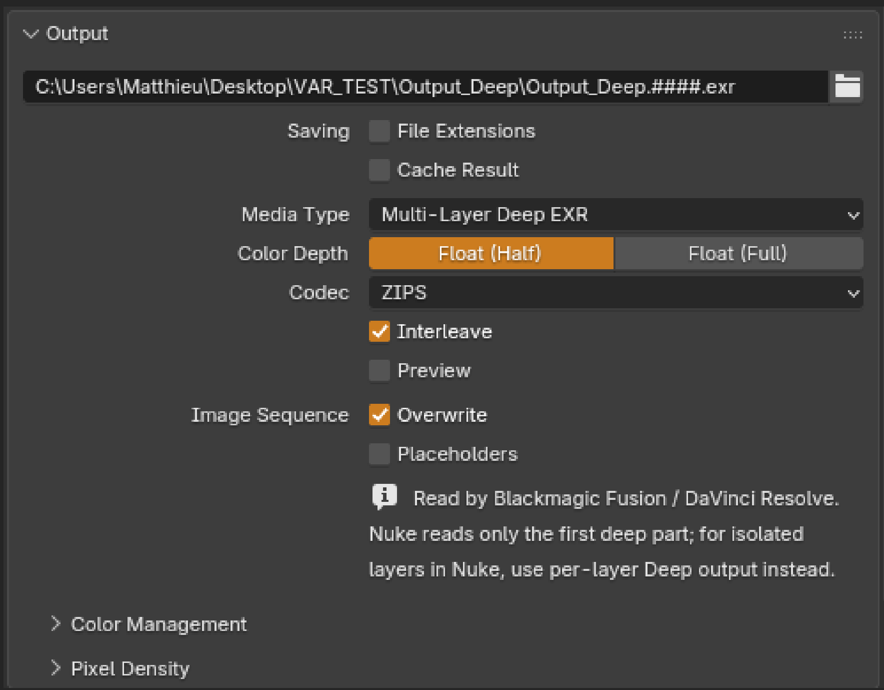


Fig. 1.6 — Selecting Multi-Layer Deep EXR in the Output panel

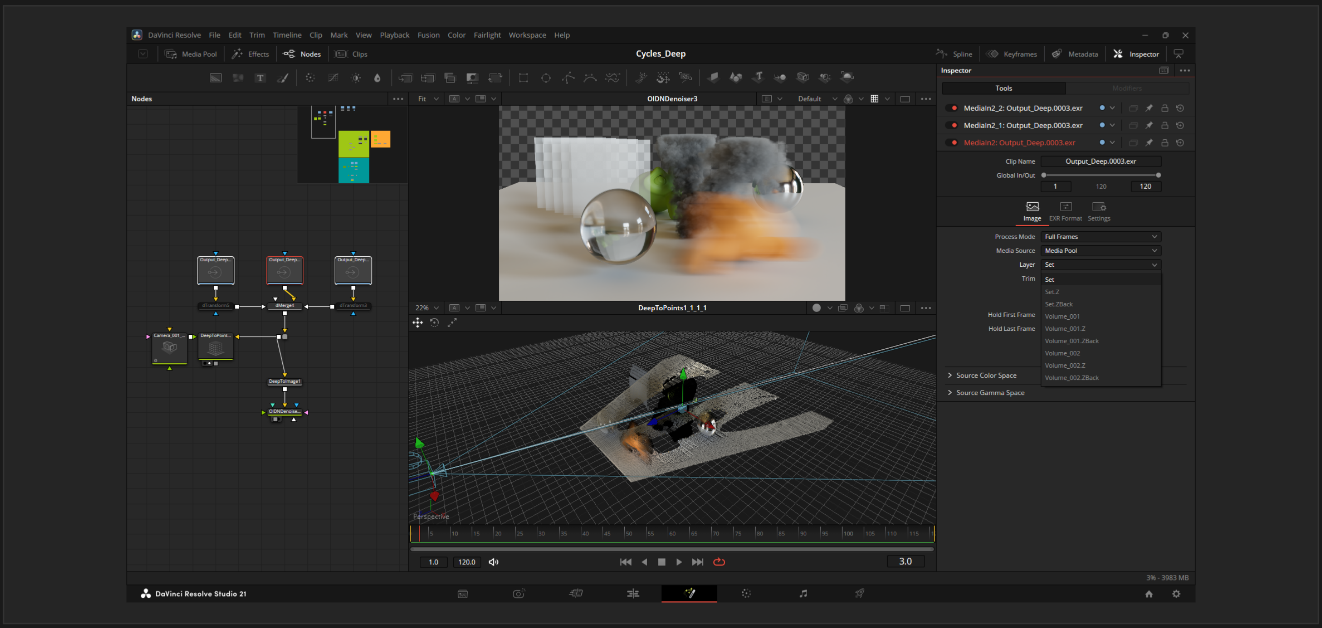


Fig. 1.7 — Multi-Layer Deep EXR read in DaVinci Resolve — Fusion

2 Z-Depth Pass

Antialiased, volume-aware and transparency-aware depth

The standard Z depth pass is upgraded to an **antialiased, volume-aware and transparency-aware** depth. There is no mode selector — just enable **Depth**, and it works with or without Deep EXR.

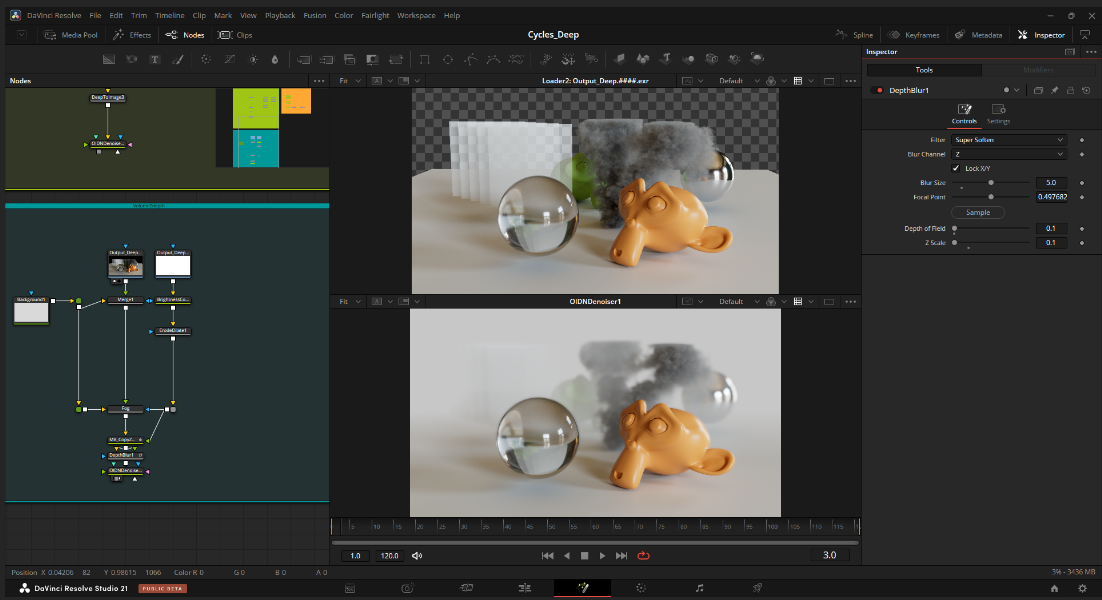


Fig. 2.1 — New Z-Depth with volumes, viewed in DaVinci Resolve — Fusion

2.1 Overview

The final **Depth** channel is a **min(surface, volume)** combine — *nearest wins*, like a compositor "Darken". This is immune to the over-bright that an alpha-over depth produces under motion blur and depth of field.

> **Difference vs stock Blender**
> *This diverges from stock Blender's Z (non-antialiased, single sample, 1e10 on empty pixels). The output here is a true antialiased depth, ready for Normalize, defocus, and depth-driven atmospherics.*

> **CPU, CUDA & OptiX — SVM and OSL**
> *The antialiased volume-aware Z-Depth runs on CPU, CUDA and OptiX, with both SVM and OSL volume shaders on GPU. On OptiX, OSL volume shading is resolved through the OSL shade pipeline, so the volume depth is computed natively (no surface-only fallback).*

2.2 What the Pass Does

- **Antialiased surface depth** — accumulated per sample and coverage-weighted, so edges are smooth instead of the hard, aliased single-sample Z of stock Blender. The value is the true (unpremultiplied) scene depth.
- **Volume aware** — volumes contribute a per-pixel depth (jittered sub-rays for clean silhouettes, with Depth of Field and Motion Blur), composited so the nearest of surface / volume wins. Multiple and overlapping volumes are handled correctly.
- **Opacity-weighted volume depth** — low-opacity volumes (wispy smoke, thin atmosphere) are blended proportionally toward the far background depth rather than reading at their full near depth. A fully opaque volume reads at its true near depth; a nearly transparent volume reads near the background.
- **Transparency aware** — semi-transparent surfaces (and antialiased edges) over the background read at a weighted depth blended toward the far background, so a depth-driven fog / atmosphere shows correctly through them.
- **Clean background** — empty / transparent pixels are filled with the maximum captured scene depth, so a Normalize node maps the whole frame cleanly (no 1e10 spike on empty pixels like stock Blender).

2.3 Limitations

Motion blur over volumes. A motion-blurred surface partially in front of a volume can produce a hard depth edge at the motion-blur boundary, because the surface and volume depths are combined per pixel after each is resolved.

> **Workaround**
> *Render the surface and the volume on two separate View Layers, each with its own Depth pass, then combine them with a Math > Minimum node in the compositor. Each pass is then computed in its own clean context and the motion blur stays correct.*

Surface / volume boundary. A thin gap or hard edge may be visible at the contact line between a surface and an adjacent volume. A one-pixel **Erode / Dilate** in the compositor is enough to clean it up in most cases. The two-View-Layer workaround above also eliminates it.

3 GPU Render Options

Direct render-device access and the OSL group-data budget

Two GPU render conveniences are surfaced directly in **Render Properties**: pick your render devices without opening Preferences, and raise the per-shader OSL group-data budget when heavy OSL graphs need it.

3.1 Render Devices in the Device Panel

The Cycles **Device** panel (Render Properties) now shows the render devices directly when GPU is selected — the backend (CUDA / OptiX / HIP / oneAPI) and the per-device checkboxes, mirrored from Preferences. Switch backend or choose which GPUs to render on without leaving the render settings.

3.2 OSL Group-Data Budget

Open Shading Language shaders allocate a fixed per-thread *group-data* buffer on GPU, capped at **2048 bytes** in stock Blender — heavy custom OSL graphs can exceed it and fail to load with a "group data size" error. This build adds an **OSL GPU Group Data** selector in **Render Properties** (shown when **Open Shading Language** is enabled): from **2048** (default) up to **6144** bytes, in 1024-byte steps. Raising it lets heavier custom OSL graphs compile and render on GPU.

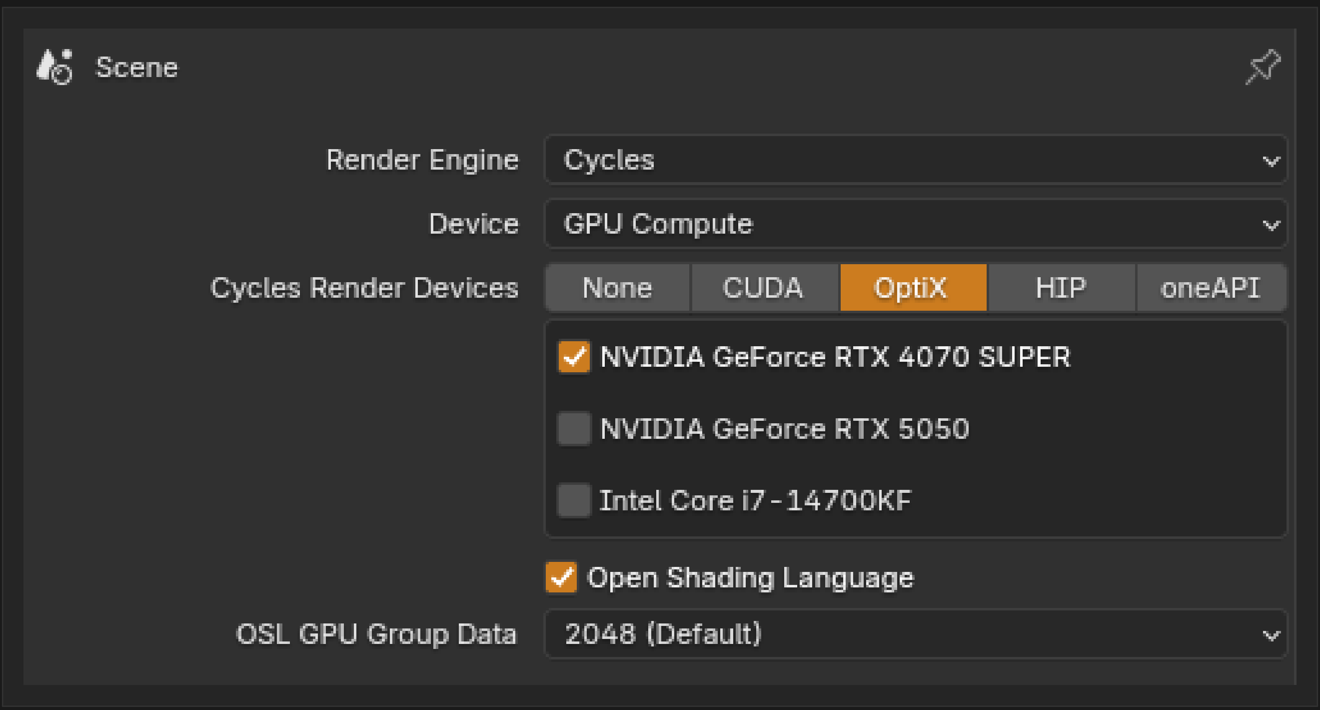


Fig. 3.1 — Render devices and the OSL group-data budget in Render Properties

3.3 Trade-off & Constraints

> **Trade-off**
> *The buffer is reserved per-thread, so a larger budget lowers GPU occupancy and slows down all OSL shading in the render — the impact can be significant. It only affects GPU rendering; OSL on CPU is unlimited. Changing the value recompiles the OSL kernels.*

If a shader still exceeds the highest budget, simplify the OSL graph (fewer layers, break up node groups, drop large arrays) or render that shader on CPU, where group data is unlimited. The **CYCLES_OSL_GROUPDATA_ALLOC** environment variable overrides the menu for advanced setups.

4 Environment Variables in File Paths

Portable, pipeline-friendly path resolution

File paths anywhere in Blender — render output, textures, libraries, caches, and even Preferences — now support **environment variable expansion**. This makes .blend files portable across machines and render farms with different local configurations.

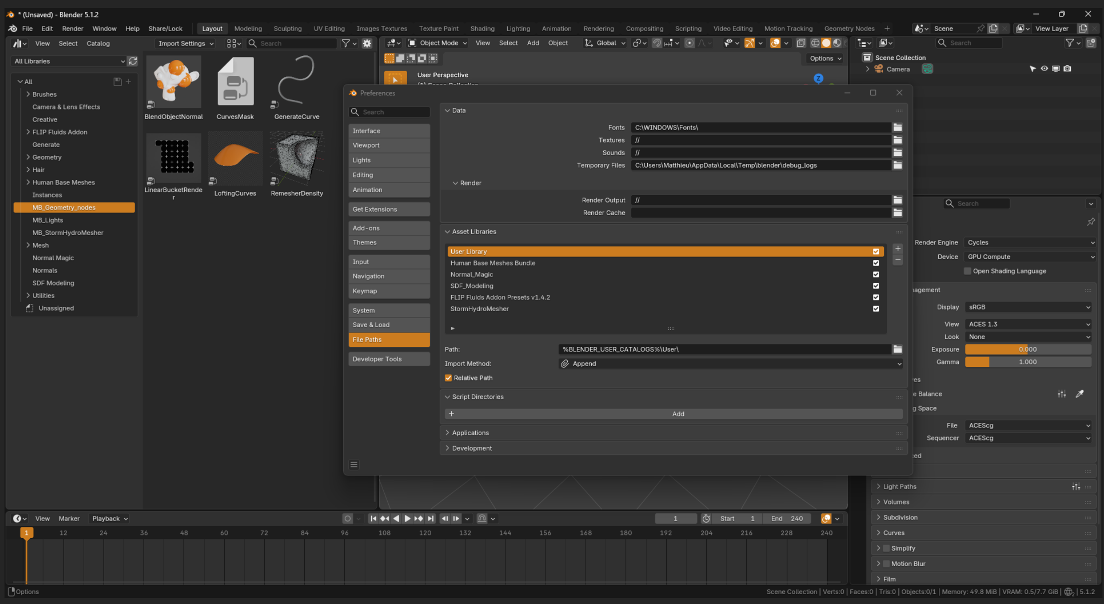


Fig. 4.1 — Full environment-variable support across Blender file paths

4.1 Overview & Supported Syntaxes

All three common syntaxes are supported, cross-platform:

```
$MY_PROJECT/renders/####.exr
${SHOT_DIR}/textures/diffuse.png
%PIPELINE_ROOT%/assets/char.blend
```

Variables are expanded at **path resolution time**, not at load time, so a .blend file stays portable: the same scene resolves to the right local paths on each artist's machine and on the farm, driven entirely by the environment.

> **Tip**
>
> *Combine environment variables with Blender's // relative-path notation and #### frame padding for fully pipeline-driven, machine-independent output paths.*

5 Qt Tools (PySide6)

An extensible, separate-process Qt UI framework for pipeline tools

This build bundles a small framework to run rich **Qt (PySide6)** user interfaces for pipeline tools — asset browsers, farm submitters, advanced control panels — beyond what the native Blender UI allows.

5.1 Overview

- **Separate process** — each Qt window runs in its own process, fully isolated from Blender. Dragging or resizing the Blender window never freezes the tool (and vice-versa).
- **Always on top** — tool windows stay above Blender so they are never lost behind it.
- **Live two-way sync** — editing a value in the tool updates Blender instantly; changes made in Blender flow back to the tool.
- **Auto-loaded bridge** — the headless **MattRM2 Qt Bridge** add-on loads automatically; it relays data between Blender and the tools and has no UI of its own.

5.2 Enabling the Deep EXR Demo

A Deep EXR control panel ships as a demo of the framework. To use it:

1. In **Preferences > Add-ons**, enable **MattRM2 — Deep EXR Panel (Qt demo)** (search "MattRM2").
2. In the 3D Viewport, open the **N-panel** and go to the **MattRM2 Qt** tab.
3. Click **Open Deep EXR Panel** — the Qt window opens, live-bound to the Deep EXR render settings.

5.3 Building a Tool — for Developers

A control-panel tool is essentially a layout of Blender property paths. The shared SDK builds the right widget per type and handles all the IPC and theming:

```
from mattrm2_qt_sdk import run_control_panel

run_control_panel("My Tool", layout=[
  ("Output", [
    ("Codec", "scene.render.deep_exr_output_compression"),
  ]),
])
```

Architecture:

- **Bridge** (auto-loaded, Blender side) — a stable public API: *launch_tool()*, *register_handler()*, *register_tools_dir()*. It resolves context-relative RNA paths (e.g. *scene.render.codec*) and relays get/set.
- **SDK** (Qt side) — the shared charter: Matt Dark theme, always-on-top window, auto-built widgets from the schema, two-way sync. Tools import it.
- **Where code lives** — UI and third-party integrations (render farm, asset / production trackers) live in the Qt tool process (its own Python, free to add dependencies); custom Blender-side logic is registered as a handler on the bridge.

License

Qt for Python (PySide6) and Qt are bundled under the GNU LGPL v3 (dynamically linked). The license texts are in the distribution's licenses/ folder.

6 View Layer Attribute Overrides

Maya-style per-render-layer overrides — up to the render engine

Override almost **any property per view layer** — the workflow of Maya's Render Setup, native in Blender. Right-click a property, choose **Add Layer Override**, and that layer gets its own value; every other layer keeps the base one. The system is a dedicated per-layer store (separate from Library Overrides), applied on each layer's evaluated data — the original scene data is never touched.

6.1 Overview & Workflow

- **Add / Remove** — right-click any supported property: object and light/camera data attributes, shader node sockets, and render settings. *Add Layer Override* captures the current value for the active view layer.
- **Visual state** — overridden properties show a **bold orange label** (color configurable in Preferences > Themes > User Interface > State); checkboxes highlight their own text.
- **Per-layer editing** — with an override active, editing the property (fields, color picker, eyedropper, copy/paste, hex) changes **only the active layer**. Switch layers and the UI, tooltips and material preview follow the layer's values.
- **Persistent** — overrides are saved in the .blend per view layer and restored on load.

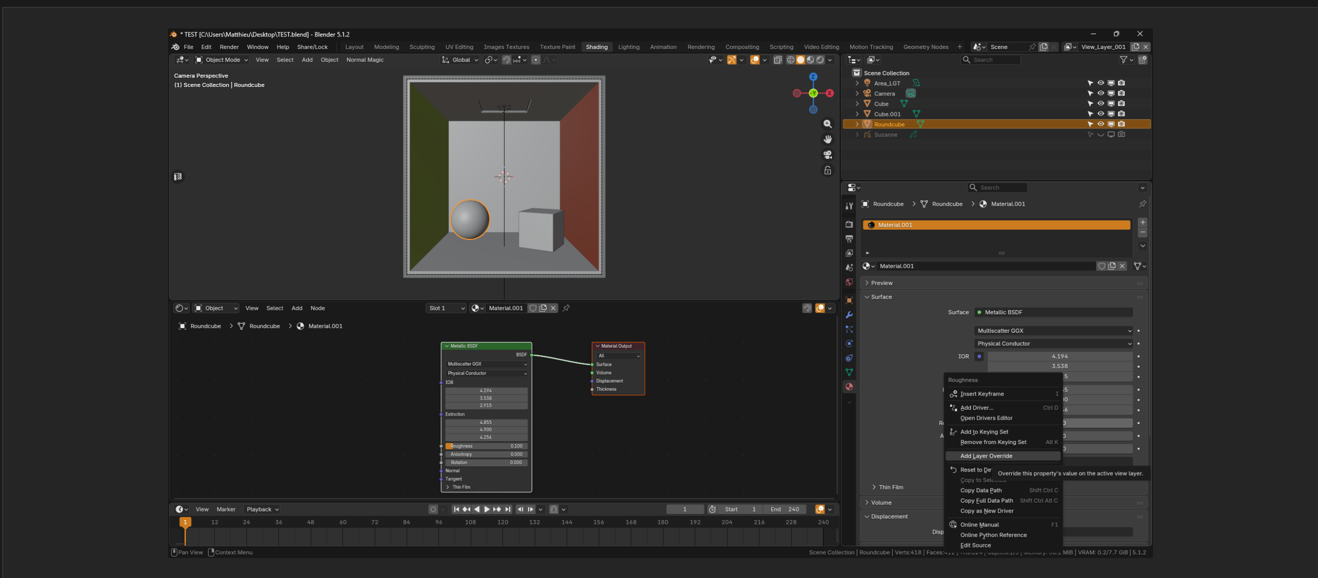
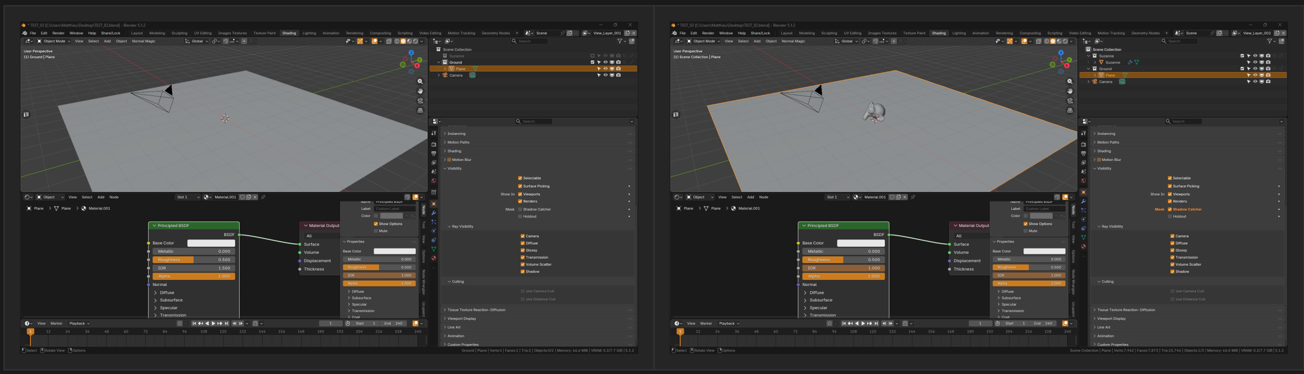


Fig. 6.1 — Right-click a property: Add / Remove Layer Override



Before — *base properties, shared by all layers*

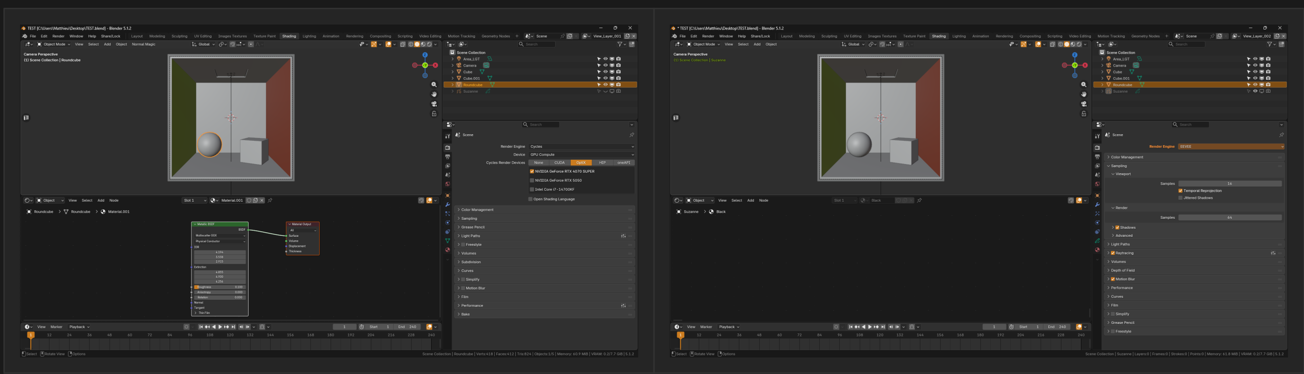
After — *per-layer overrides active: bold orange labels, layer-local values*

Fig. 6.2 — Overridden properties: bold orange labels, per-layer values.

6.2 Per-Layer Render Engine

The render engine itself can be overridden per view layer: one scene can render some layers with **Cycles and others with EEVEE in the same F12 or animation render**, into the same multilayer EXR — each layer with its own engine's pass layout.

- **Right-click the Render Engine dropdown** on the target layer and add the override, then pick the engine for that layer.
- **The whole UI follows** — render panels, view layer passes and shader node availability switch to the active layer's engine; the rendered viewport previews the active layer with its own engine.
- **Grease Pencil and Freestyle** keep compositing correctly on mixed-engine renders.



View Layer 01 — *scene engine (Cycles)*

View Layer 02 — *engine overridden to EEVEE: render UI and viewport follow*

Fig. 6.3 — Render engine per view layer: Cycles and EEVEE in one scene.

6.3 Per-Layer Samples & Shaders

- **Samples** — override *Max Samples* per layer (e.g. 4096 on the hero layer, 64 on a matte layer): each layer renders with its own count in the same render, wired into Cycles' native per-layer sampling.
- **Shader switching** — override any node socket. Overriding a *Mix Shader* factor turns it into a per-layer shader switch without duplicating materials; the material preview shows the active layer's look.

- **Engine settings** — Cycles and EEVEE settings read per layer (bounces, clamps, film transparency, motion blur toggles...), so specialized layers carry their own render configuration.

6.4 Scope & Limitations

Pipeline-geometry settings — **resolution, borders, frame range, FPS, output paths and file formats** — are deliberately not overridable: they are read once per render and keeping them per scene guarantees a consistent multilayer EXR. The right-click entry is disabled for them. Animated or driven properties cannot be overridden either (the override and the F-Curve would fight over the value).

> **Per-layer outputs**
> *For per-layer file outputs, use the compositor File Output node for now — native per-layer output paths and formats are on the roadmap.*

7 Included Bug Fixes

Upstream Cycles / UI fixes shipped ahead of official patches

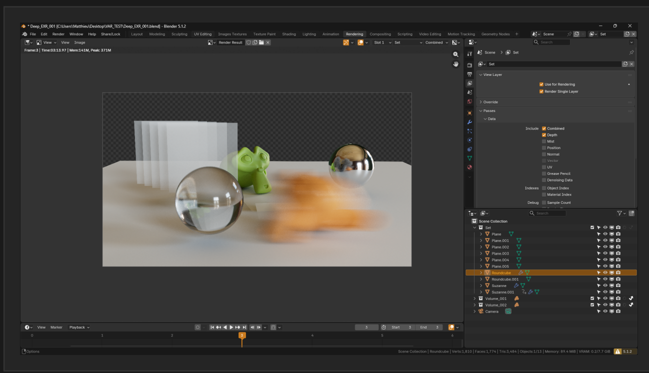
7.1 Node Editor Click-Drag

Fixed an official Blender 5.1 bug where click-dragging a node would move a different node than the one under the cursor. The incorrect node selection persisted in the .blend file. This fix is included ahead of the upstream patch.

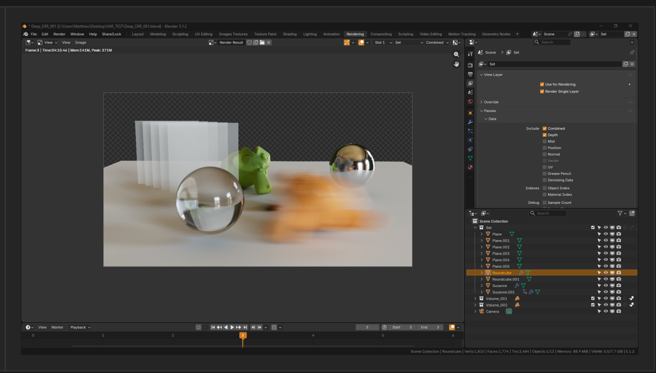
7.2 Volume Indirect-Only Shadows

Fixed a Cycles bug where volumes placed in **Indirect-Only** collections would not cast shadows — on surrounding surfaces, and on other volumes (volume-on-volume).

In particular, when the volume's bounding box intersected the floor or another object, the shadow disappeared on the intersecting surface. This build renders the shadow correctly even with large bounding-box intersections, on **CPU, CUDA and OptiX**.



Before — *stock Blender 5.1.2: no shadow where the volume's bounding box meets the floor*



After — *this build: shadow cast correctly, even on large bounding-box intersections*

Fig. 7.1 — Volume Indirect-Only shadow, before / after the fix.

> **Note**
> *The underlying issue is still present in official Blender 5.1.2 and will be reported upstream. This build ships the fix already applied.*

8 Known Limitations & Roadmap

Current constraints and what's coming next

8.1 Known Limitations

Limitation	Notes / Workaround
GPU deep uses a per-pixel layer cap	CPU is unlimited. Max Depth -1 maps to 96 on GPU; raise it in the Deep EXR panel for very dense volumes.
Multi-Layer Deep EXR: Nuke reads only the first deep part	It is a Fusion / Resolve format. For Nuke, use the standard output — one deep file per layer in Deep/<layer>/ — read individually with full isolation.
View Layer Overrides: pipeline settings are locked	Resolution, borders, frame range, FPS and output paths/formats cannot be overridden per layer (read once per render). Use the compositor File Output node for per-layer outputs — native support is on the roadmap.
View Layer Overrides: animated / driven properties	Cannot be overridden: the override and the F-Curve would fight over the value. Remove the animation first.
Heavy OSL graphs may exceed the GPU group-data budget	Raise the OSL GPU Group Data budget in Render Properties (up to 6144), simplify the shader, or render it on CPU (unlimited).
Z-Depth: motion blur over volumes	Hard depth edge where a motion-blurred surface is partially in front of a volume. Render surface and volume on separate View Layers, each with a Depth pass, and min() them in compositing.
Multi-Layer Deep EXR + multi-view	Cycles deep is mono (single view). The combined deep output is not supported with stereo / multi-view — render mono, or use the per-layer deep files.

8.2 Roadmap

Medium Term

- **DeepID** — extend deep channels with objectId, materialId, normal and albedo per fragment, and single-file per-layer isolation in Nuke.
- **Deep + DeepID Compositor Nodes** — native Blender compositing nodes for deep data manipulation (Deep Merge, Hold-Out, ID filter).
- **LPE (Light Path Expressions)** — custom AOVs via light path expressions (Arnold / RenderMan parity).

Long Term

- **Plugins for DaVinci Resolve Fusion** — DeepID Sampler, Deep Fog, Deep Relight (Nuke-workflow parity in Fusion — the 'Deep+ Tools' for Fusion).
- **Caustics** — fast, accurate caustics (Photon Map / guided raymarcher — R&D).
- **R&D for heavy production scenes.**

Feedback

This build is in active development. If you encounter unexpected behaviour, crashes, or rendering differences vs. stock Blender, please report with a .blend or minimal reproduction scene, your render settings, and your OS / GPU. Your feedback directly influences what gets built next. — Matthieu Barbié